

Queries on Grid

Siam gives an $n \times n$ integer grid to his friend Seam. Seam has to process several queries on this grid. The grid is 0-indexed, where:

- r denotes the row index
- c denotes the column index

Each query consists of three integers r, c, v . For each query, perform the following operation:

- Start from cell (r, c) and consider its value as the starting value.
- Traverse all cells that are 4-directionally connected (up, down, left, right) and have the same value as the starting cell.
- Update the value of all such connected cells to v .

Each query is applied to the grid resulting from the previous query. After processing all queries, print the final state of the grid.

Input

1. The first line contains two integers n and q ($1 \leq n \leq 10^3$, $1 \leq q \leq 10^5$) — the size of the grid and the number of queries.
2. The next n lines each contain n integers ($-10^9 \leq \text{grid}[i][j] \leq 10^9$) — representing the grid.
3. The next q lines each contain three integers: r, c, v ($0 \leq r < n$, $0 \leq c < n$, $-10^9 \leq v \leq 10^9$)

Output

Print the final grid after processing all queries. Output n lines, each containing n space-separated integers.

Example

Input

```
4 4
0 0 1 0
1 1 0 0
0 0 0 0
0 1 1 1
0 0 5
1 1 6
2 2 7
3 3 7
```

Output

```
5 5 1 7
6 6 7 7
7 7 7 7
7 7 7 7
```

Notes

1. Initial Grid State:

```
0 0 1 0
1 1 0 0
0 0 0 0
0 1 1 1
```

2. Query 1 (0 0 5):

- Target Cell: (0, 0) (Initial value: 0)
- Connected Component: {(0, 0), (0, 1)}
- Update: Recolor visited cells to 5.
- Grid State:

```
5 5 1 0
1 1 0 0
0 0 0 0
0 1 1 1
```

3. Query 2 (1 1 6):

- Target Cell: (1, 1) (Initial value: 1)
- Connected Component: {(1, 1), (1, 0)}
- Update: Recolor visited cells to 6.
- Grid State:

```
5 5 1 0
6 6 0 0
0 0 0 0
0 1 1 1
```

4. Query 3 (2 2 7):

- Target Cell: (2, 2) (Initial value: 0)
- Connected Component: {(0, 3), (1, 2), (1, 3), (2, 0), (2, 1), (2, 2), (2, 3), (3, 0)}
- Update: All 8 connected cells with value 0 are recolored to 7.
- Grid State:

```
5 5 1 7
6 6 7 7
7 7 7 7
7 1 1 1
```

5. Query 4 (3 3 7):

- Target Cell: (3, 3) (Initial value: 1)
- Connected Component: {(3, 1), (3, 2), (3, 3)} (Note: Cell (0, 2) has value 1, but is not 4-directionally connected to this component)
- Update: Recolor visited cells to 7.
- Grid State:

```
5 5 1 7
6 6 7 7
7 7 7 7
7 7 7 7
```